

A discreteness obstruction to the printed polytope approximation rate

ColdSmile
Conjecture 00000001451

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Abstract

The conjecture assigns the optimal facet count the rate $c_d \varepsilon^{(d-1)/2}$. Both source languages print this positive exponent. In dimension two, no natural number can lie between two fixed positive multiples of $\sqrt{\varepsilon}$ for every sufficiently small positive ε . Consequently the conjecture fails as an exact formula, a positive leading asymptotic, and even a positive two-sided order estimate. The Lean formalization retains the actual Euclidean volume approximation condition, geometric facets, and attained minimum.

1 Statement and scope

Let B_d be the closed unit ball in Euclidean $\mathbb{R}^d$. The source calls a polytope P an ε -approximation when

$$\text{vol}(B_d \triangle P) \leq \varepsilon \text{vol}(B_d).$$

It states that the number of facets of the minimal approximating polytope is $c_d \varepsilon^{(d-1)/2}$, with an explicit exact constant (the constant in the optimal approximation order). The unmodified English and Chinese text is supplied in `ORIGINAL.md`.

We interpret “minimal” as minimizing the number of facets. The phrase “is” allows a literal equality, whereas “optimal approximation order” suggests an asymptotic statement. We address both, as well as the weaker positive two-sided order. In each reading a leading or optimal-order constant is finite, strictly positive, and independent of ε . It suffices to refute dimension $d = 2$, even if a formula is required only on an arbitrarily small interval $0 < \varepsilon < \delta$.

Theorem 1. *There do not exist $a, b, \delta > 0$ such that for every $0 < \varepsilon < \delta$ an attained minimum facet count n satisfies*

$$a\sqrt{\varepsilon} \leq n \leq b\sqrt{\varepsilon}.$$

In particular the printed exact formula and positive leading asymptotic are false in dimension two, and hence the dimension-uniform conjecture is false.

No sign-corrected rate $\varepsilon^{-(d-1)/2}$ is asserted or refuted here. A zero constant is not a positive leading or optimal-order constant. We also make no claim about a bare one-sided upper-bound formulation. These are distinctions of scope, not alterations of the printed exponent.

2 Geometric definitions and attainment

The formal ambient space is `EuclideanSpace ℝ (Fin d)`. A polytope is a convex hull $P = \text{conv}(S)$ of a finite set of points. A facet is a nonempty proper exposed face F of P with

$$\dim \text{aff}(F) + 1 = \dim \text{aff}(P).$$

These are intrinsic dimensions. In the full-dimensional case they give the usual codimension-one facets. The definition permits all convex polytopes, without imposing containment in or around the ball, symmetry, or a prescribed vertex arrangement. Volume is the standard Lebesgue volume on Euclidean space; symmetric difference is the actual set operation. For positive ε , the nonnegative extended-real factor used in Lean is precisely ε .

The relation `HasFacetCount P n` means that $\{0, \dots, n-1\}$ is in bijection with the set of these geometric facets. It is not a totalized cardinality of an infinite set. The relation `IsMinimumFacetCount d ε n` says that an actual approximating polytope has n facets and every admissible facet count m satisfies $n \leq m$.

Lemma 1. *Every polytope has finitely many exposed faces and hence finitely many facets. Its finite facet count is unique. For each error, an attained minimum facet count exists if and only if an approximating polytope exists; that minimum is unique.*

Proof. Write $P = \text{conv}(S)$ with S finite. The set P is compact. Every exposed face F is compact and convex, and its extreme points are among the extreme points of P , which are among S . By the finite-dimensional Krein–Milman theorem, $\overline{F} = \overline{\text{conv}(\text{ext } F)}$. Thus exposed faces belong to the image of the finite power set of S under $A \mapsto \text{conv}(A)$. Facets form a subset of this finite set. Two bijections with finite ordinal sets force their natural-number cardinalities to agree.

If an approximating polytope exists, its finite facet count makes the set of admissible natural counts nonempty. Its least element is an admissible count, so a polytope attaining it exists. Conversely an attained minimum includes such a polytope by definition. Two minima satisfy both opposite inequalities, so are equal. $\square$

All assertions in Lemma 1 are formalized. The disproof does not need a separate construction of approximants at every error: each of the three asserted rates already requires actual minimum witnesses at sufficiently small errors. We prove that no such witnesses can satisfy the printed rate, even before using their additional geometric properties.

3 The integer obstruction

Fix arbitrary $a, b, \delta > 0$ and set

$$t = \frac{1}{2} \min\{\sqrt{\delta}, 1/b\}, \quad \varepsilon = t^2.$$

Then $t > 0$, $t < \sqrt{\delta}$, and $t < 1/b$. Hence $0 < \varepsilon < \delta$, $\sqrt{\varepsilon} = t$, and $bt < 1$. If $n \in \mathbb{N}$ satisfied the two bounds in Theorem 1, then

$$0 < at \leq n \leq bt < 1.$$

There is no natural number strictly between 0 and 1. This proves Theorem 1. The obstruction covers every natural count; there is no numerical search or finite sampling of errors.

4 The three readings of the conjecture

For $d = 2$ the printed real exponent is $(2 - 1)/2 = 1/2$, so the rate is $\sqrt{\varepsilon}$ exactly.

Exact formula. An equality $n = c\sqrt{\varepsilon}$ with $c > 0$ implies the two-sided bounds with $a = b = c$, contradicting Theorem 1.

Positive leading asymptotic. Suppose the minimum counts satisfy $n(\varepsilon)/(c\sqrt{\varepsilon}) \rightarrow 1$ as $\varepsilon \rightarrow 0^+$, for $c > 0$. Choose the tolerance $1/2$ in the ordinary epsilon-delta definition. At all sufficiently small positive errors the ratio lies between $1/2$ and $3/2$, and therefore

$$(c/2)\sqrt{\varepsilon} \leq n(\varepsilon) \leq (2c)\sqrt{\varepsilon}.$$

Theorem 1 again contradicts the claim. The formal predicate quantifies actual minimum witnesses relationally for every tolerance; Lemma 1 makes their count unique wherever they exist. The auxiliary module also proves the corresponding obstruction for Mathlib’s standard filter-based asymptotic equivalence of functions.

Optimal two-sided order. An order $\Theta(\sqrt{\varepsilon})$ with positive fixed lower and upper constants is exactly the already-refuted pair of bounds. Thus the disproof does not rely solely on literal equality of an integer with a noninteger. It rules out the natural optimal-order interpretation too. Since no positive constant has the claimed behavior, asking for that constant to be explicit cannot restore the statement.

5 Lean correspondence and reproduction

`Geometry.lean` defines the geometric objects and proves finiteness, unique counts, and the equivalence between minimum attainment and existence of an approximant. `Vanishing.lean` proves the integer obstruction for every positive choice of constants and error interval, including the standard asymptotic-equivalence version.

`Solution.lean` defines all three readings using attained geometric minimum witnesses. The final universal negations in namespace `TLMC1451` are:

```
conjecture_exact_false
conjecture_asymptotic_false
conjecture_order_false.
```

Each follows from a corresponding dimension-two theorem; the universal domain $d \geq 2$ contains that counterexample dimension.

The project pins Lean 4.19.0 and Mathlib v4.19.0, with exact dependency revisions in its manifest. From the supplied `lean/` directory, use the supplied manifest to obtain dependencies and the Mathlib cache with `lake exe cache get`, then run `lake build`. The supplied verification script additionally replays every authored module with warnings treated as errors and inspects compiled axiom dependencies. See `VERIFICATION.md` for the recorded commands, source identities, and audit scope. The mathematical argument is symbolic; it relies on no auxiliary numerical computation. Local verification and independent pre-submission review do not constitute maintainer acceptance.